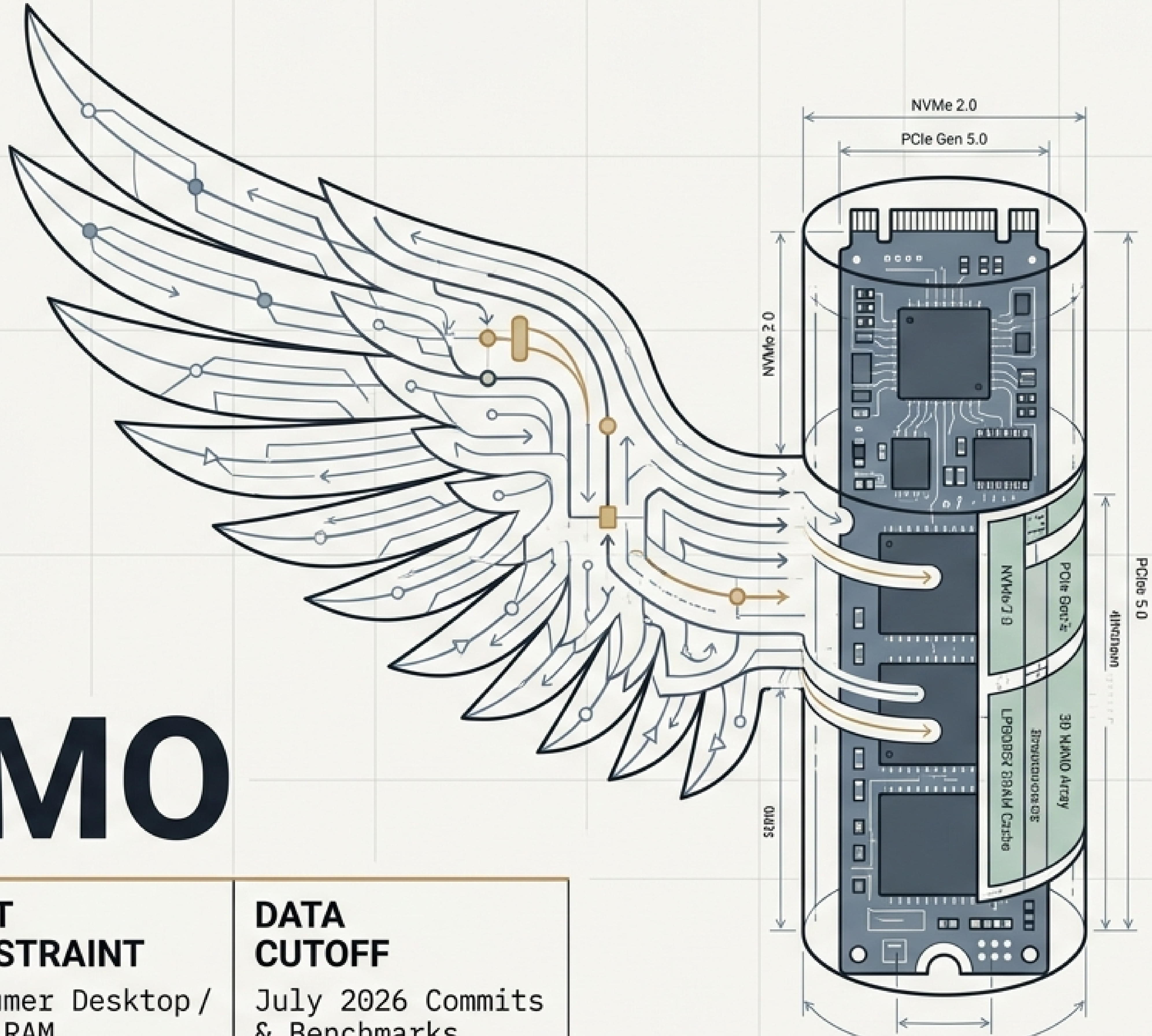


PENG-MIMO

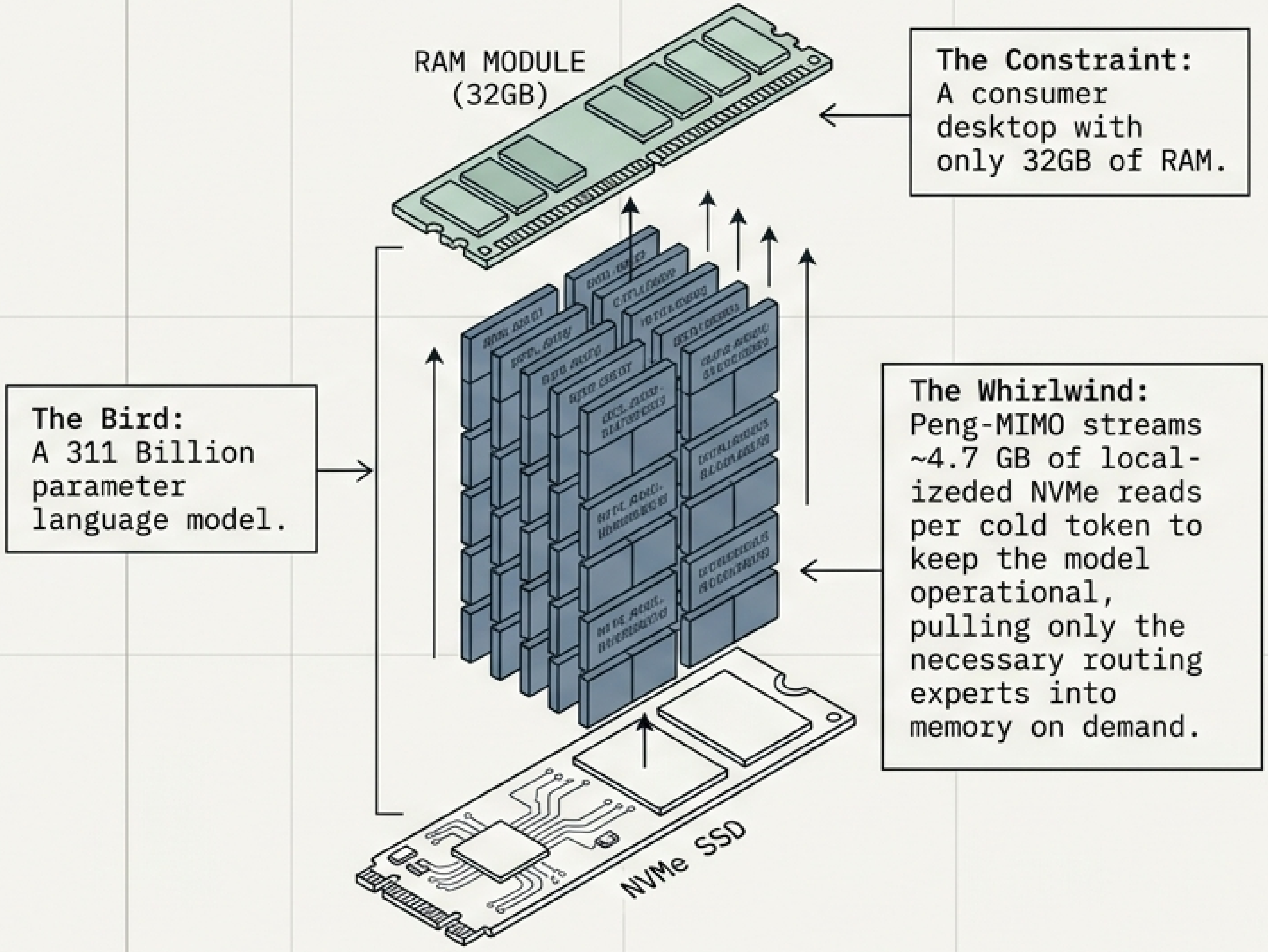
CORE ENGINE	TARGET ARCHITECTURE	HOST CONSTRAINT	DATA CUTOFF
Colibri-style streaming MoE	MiMo-V2.5 / 311B Parameters	Consumer Desktop / 32GB RAM	July 2026 Commits & Benchmarks

Riding the Digital Whirlwind

The Myth

In the *Zhuangzi*, the mythological Peng bird is a creature of immense size. It cannot fly through normal aerodytics; it stays airborne solely by riding a colossal whirlwind.

The Engineering Reality



Maximizing Colibri Architecture Reuse While Respecting Physical Constraints

Metric	MiMo-V2.5	GLM-5.2	Hy3	Qwen3.5-122B
Disk Size (Int4)	~165 GB (Fits)	~370 GB (Fails)	~157 GB	~65 GB
Active Params / Token	15B	40B	21B	10B
Router Type	Identical to Colibri	Identical	Different	Different
Attention Mechanism	Simple GQA	MLA+DSA	GQA (Hetero)	Gated DeltaNet
Checkpoint Format	FP8 128x128	-	Different	Different

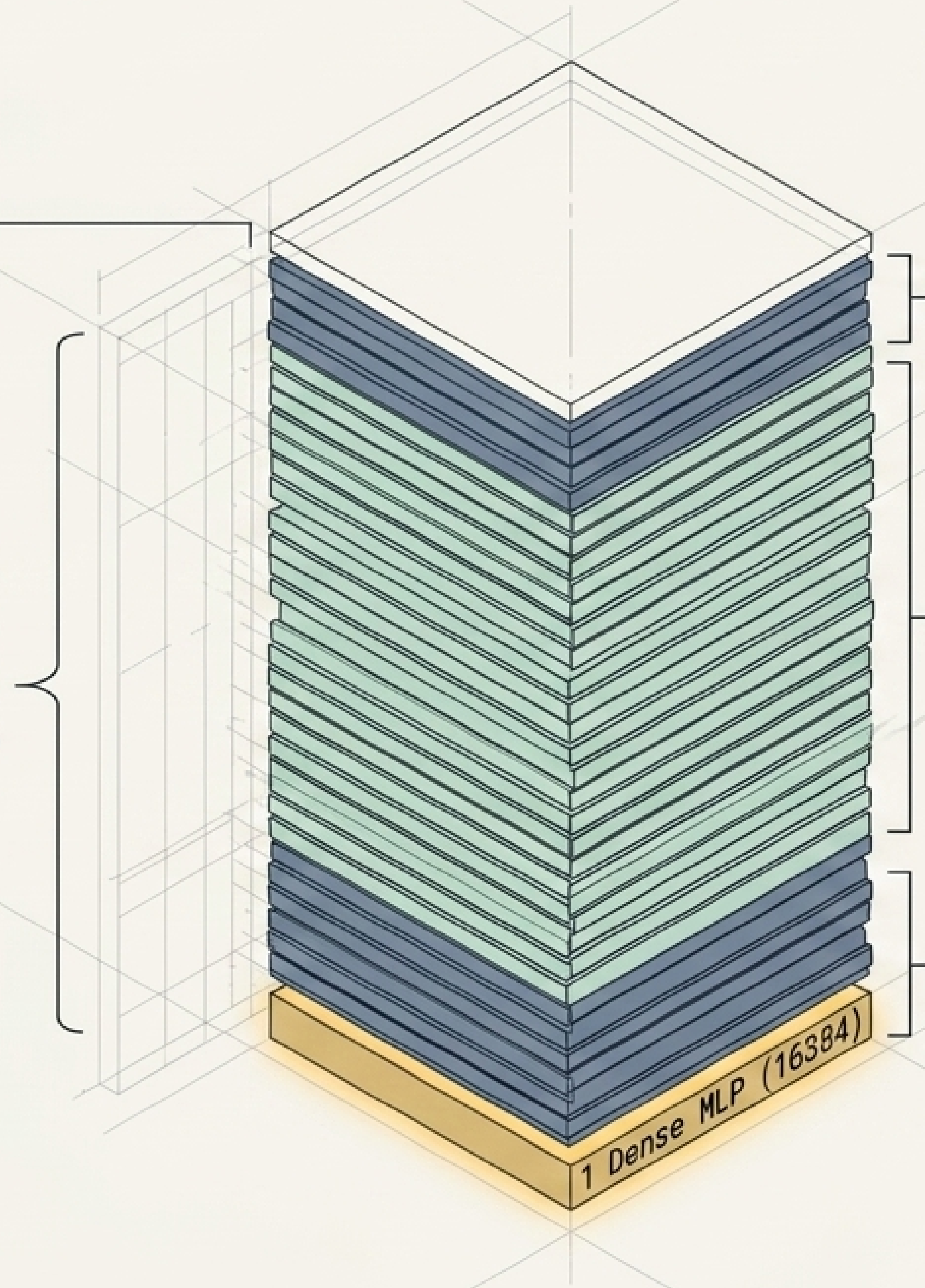
Perfect footprint. At 165GB total and only 15B active parameters, it fits the consumer NVMe ceiling perfectly without requiring a complex C-based router rewrite.

The Logical Geometry of MiMo-V2.5

Vocabulary Space

152,576 byte-BPE style tokens.

48 Total Layers:
1 Dense MLP (16384),
47 MoE layers
(256 experts top-8).



Hybrid Attention Breakdown

9 Full Layers: 64Q/4KV heads. Sees total history.

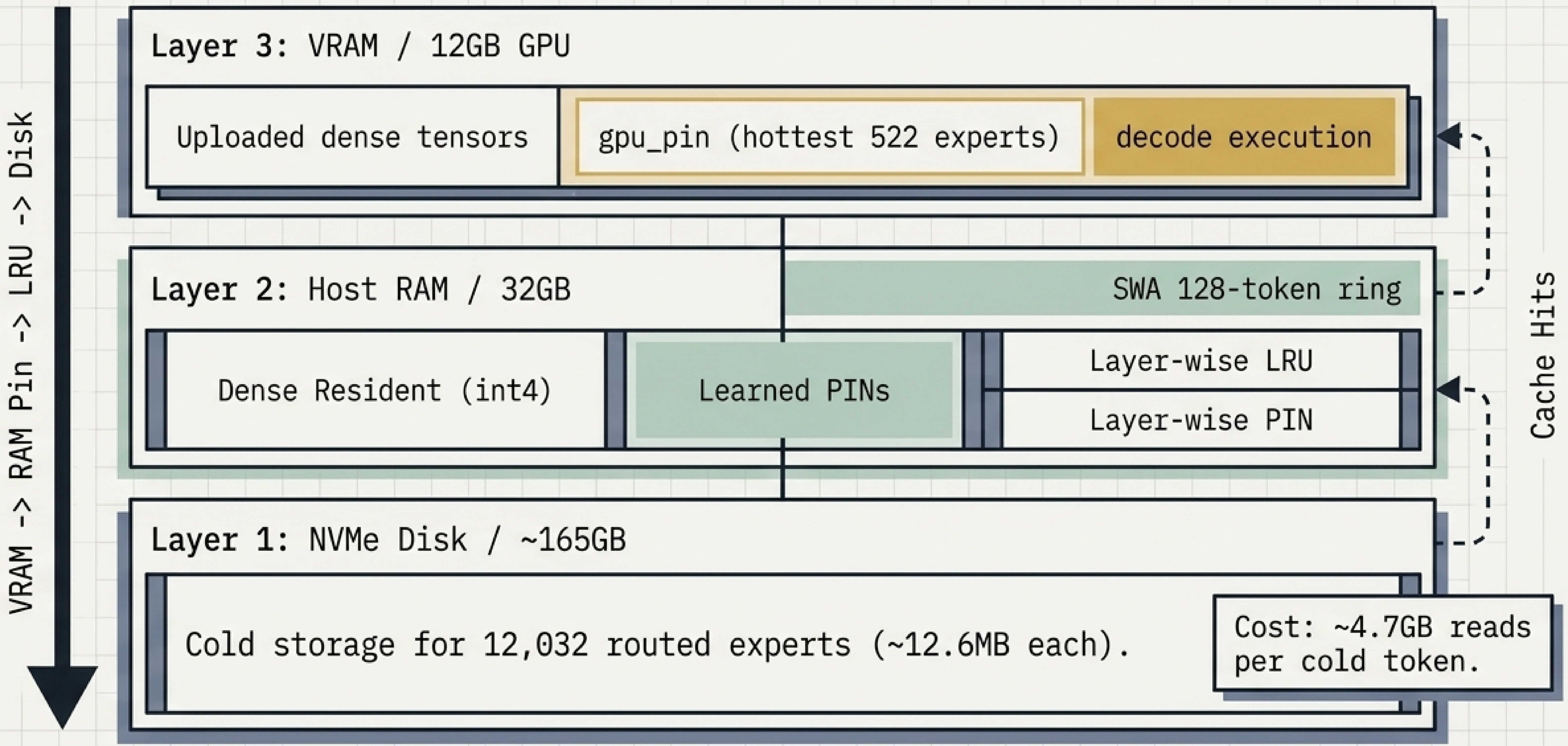
39 SWA Layers: Sliding-Window 128 context (64Q/8KV heads) + attention sink bias.

Crucial Advantage: 39 out of 48 layers only need a KV-cache of 128 tokens, making long-context process incredibly cheap in host RAM.

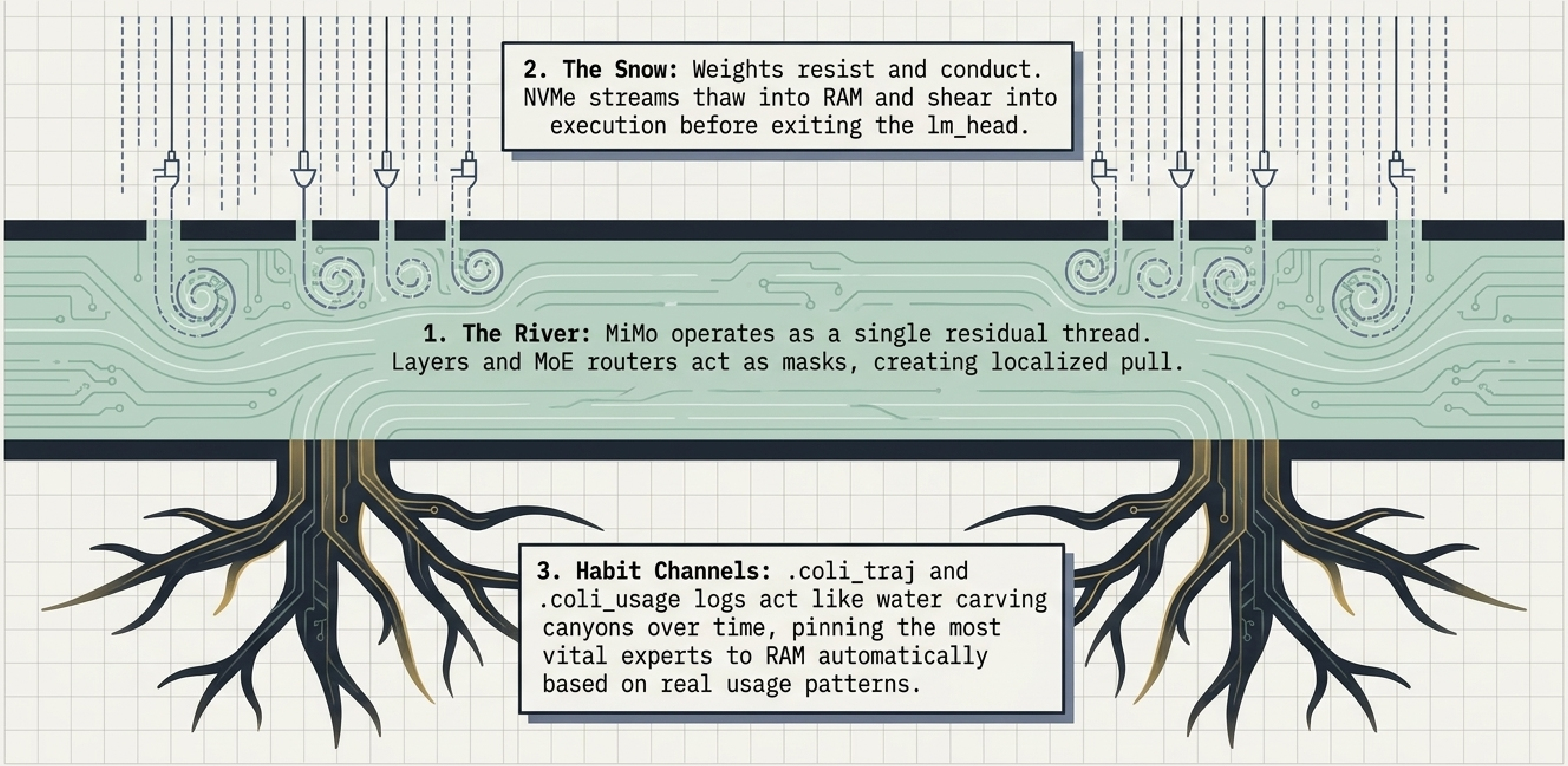
Engine Nuance

[x] Fused QKV tensor
[x] V scaled x0.707 before attention
[x] FP8 e4m3 checkpoint format

The Physical Lookup Priority

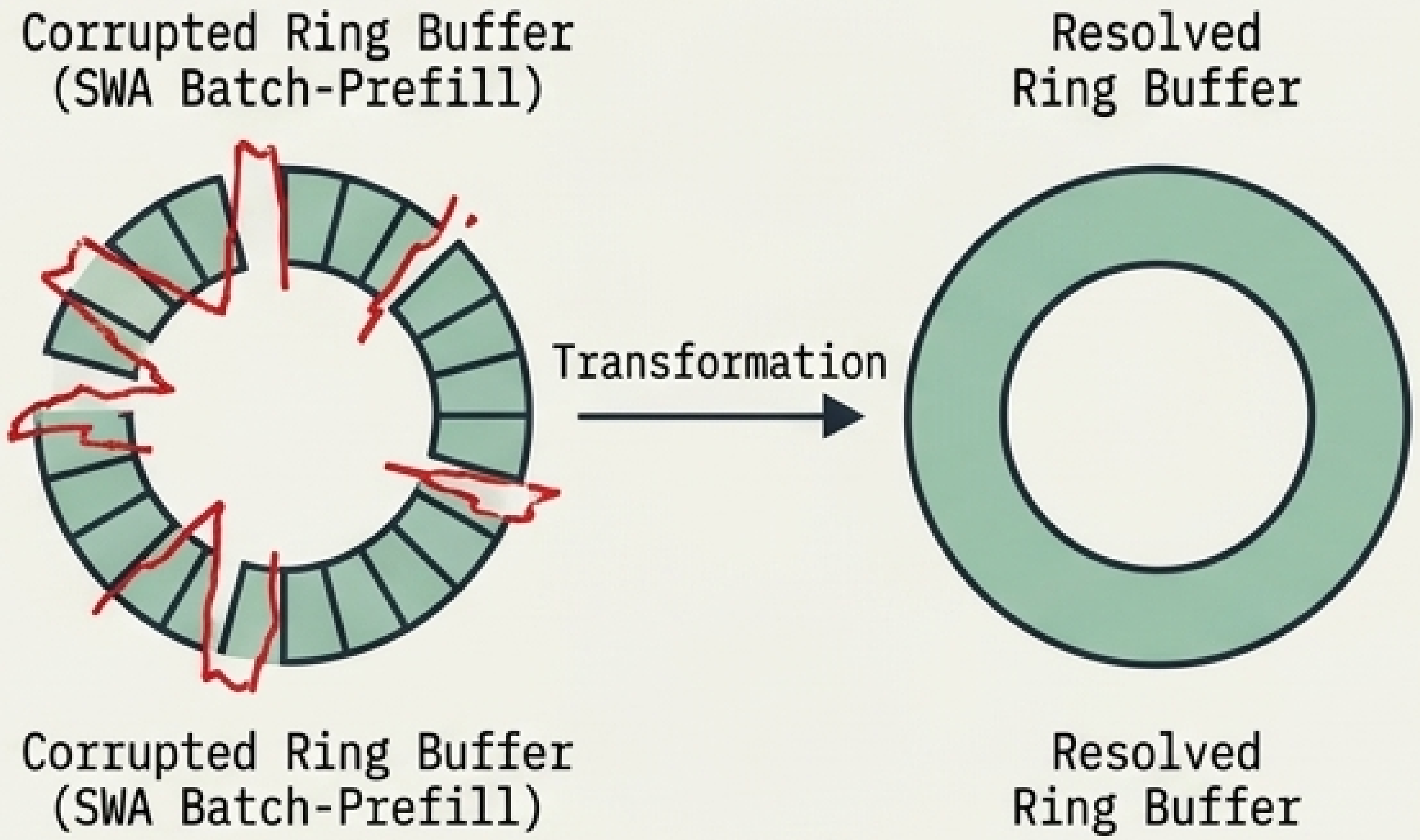


‘Corriente Peng’: The Residual River



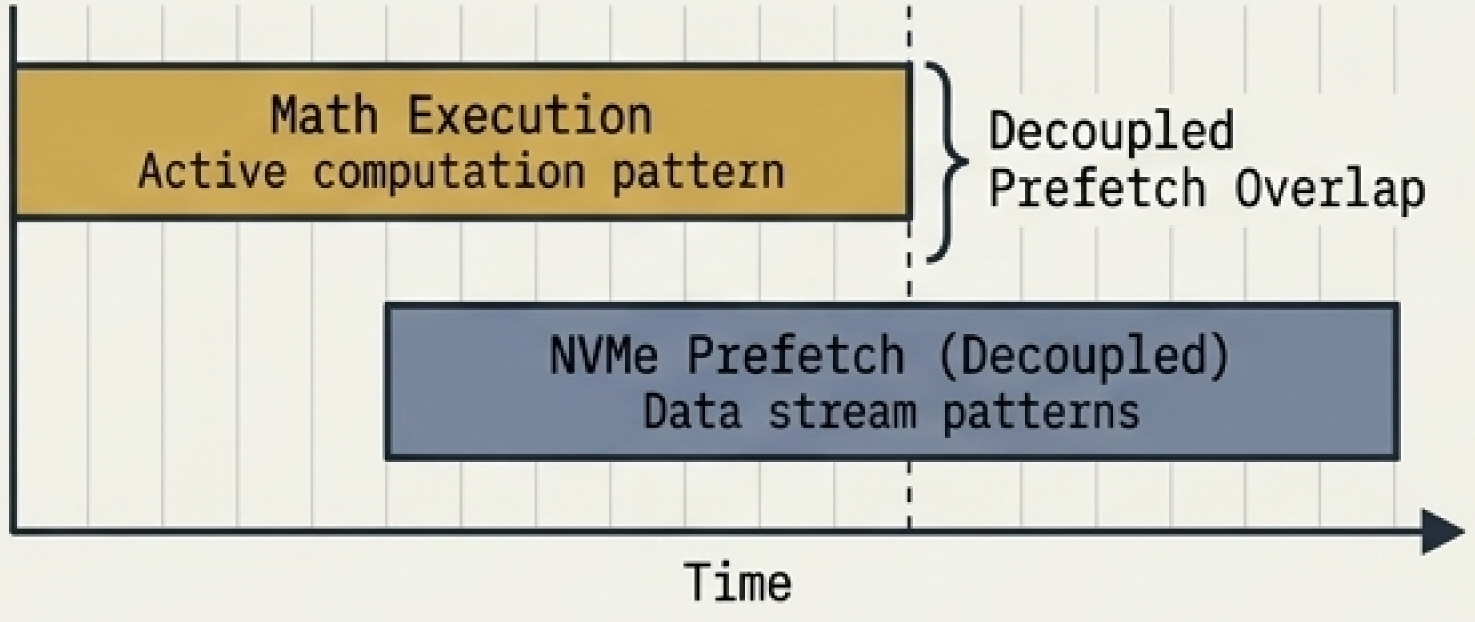
July 2026 Core Commits & Optimizations

mimo: fix SWA ring batch-prefill corruption



Impact: Resolved memory corruption in Sliding Window Attention ring buffers during the critical batch prefill phases.

mimo: PILOT/LOOKA prefetch + RoPE table



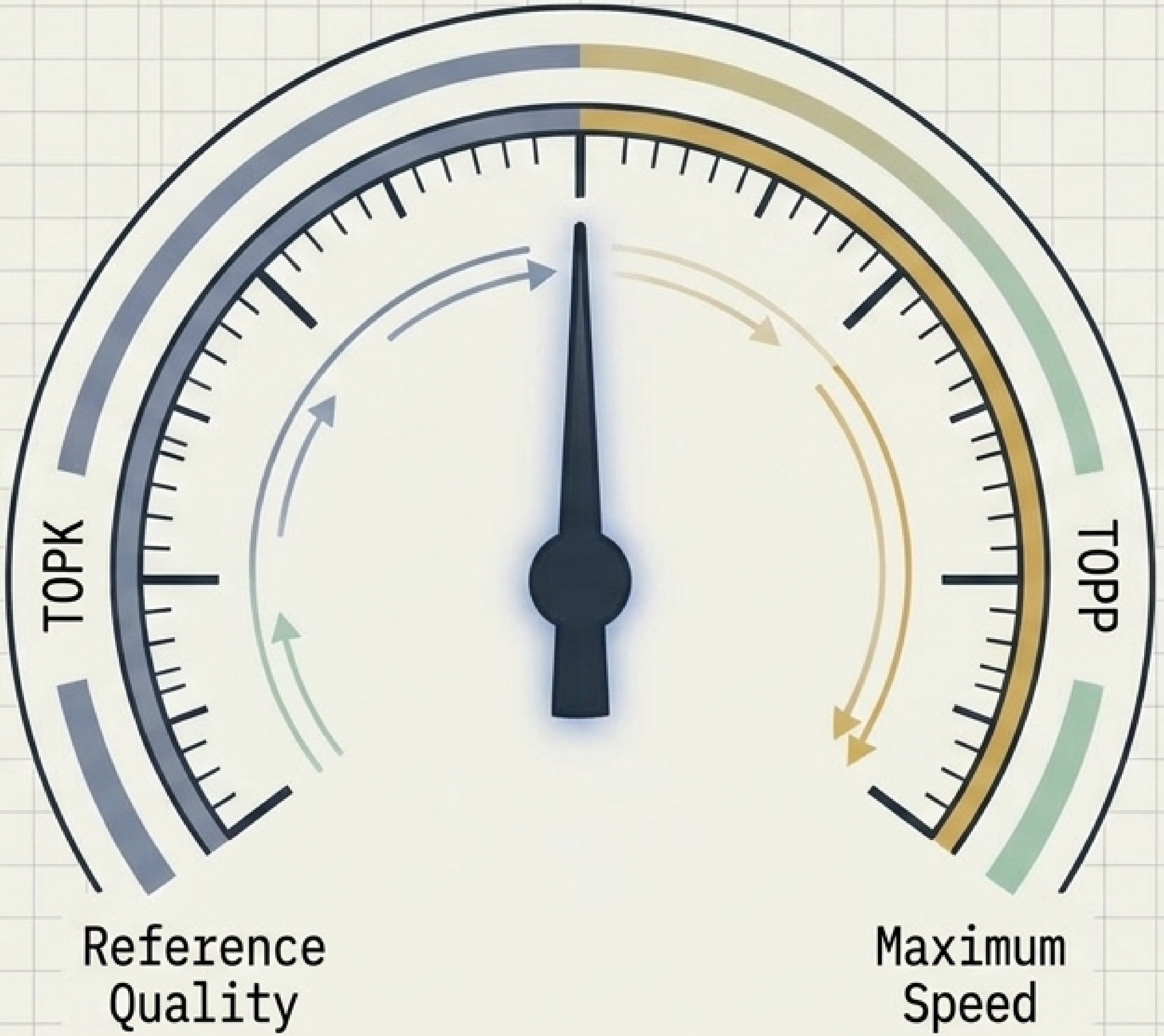
Mechanism: Overlaps disk I/O by decoupling expert prefetching from math execution.

The Result: At RAM-starved limits (cap->3), PILOT yields a massive **~1.3x to 2.0x speedup** purely by hiding NVMe latency in the background.

The Rule of the Frontier: Nothing is Free

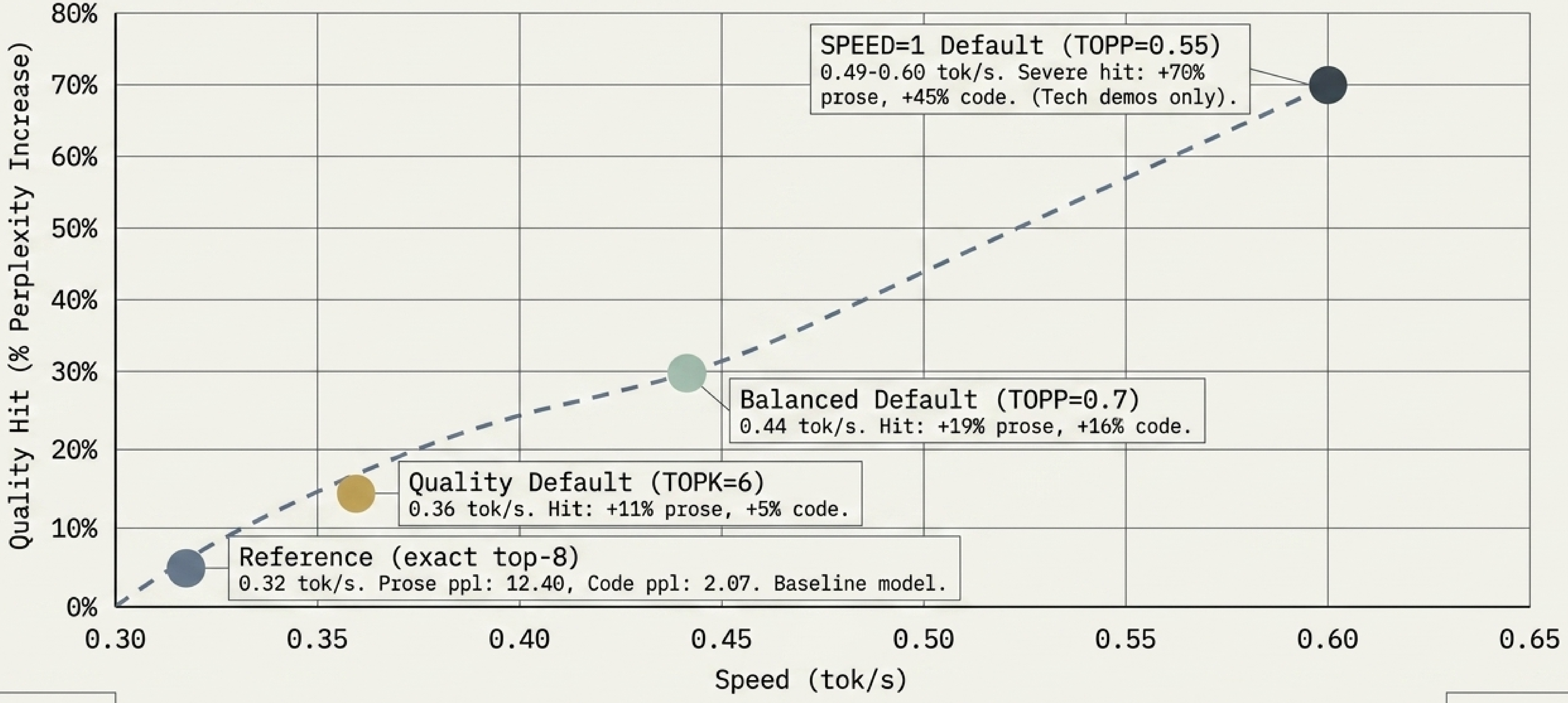
To push beyond hardware physics, Peng-MIMO utilizes expert trimming knobs.

The Iron Law: Generation speed tracks the volume of experts loaded almost linearly.



Methodology Note: All measurements on the frontier (July 13th) were evaluated via teacher-forcing perplexity on a fixed 1,534-token prose/code corpus to objectively measure the exact cost of speed.

Mapping the Quality vs. Speed Frontier

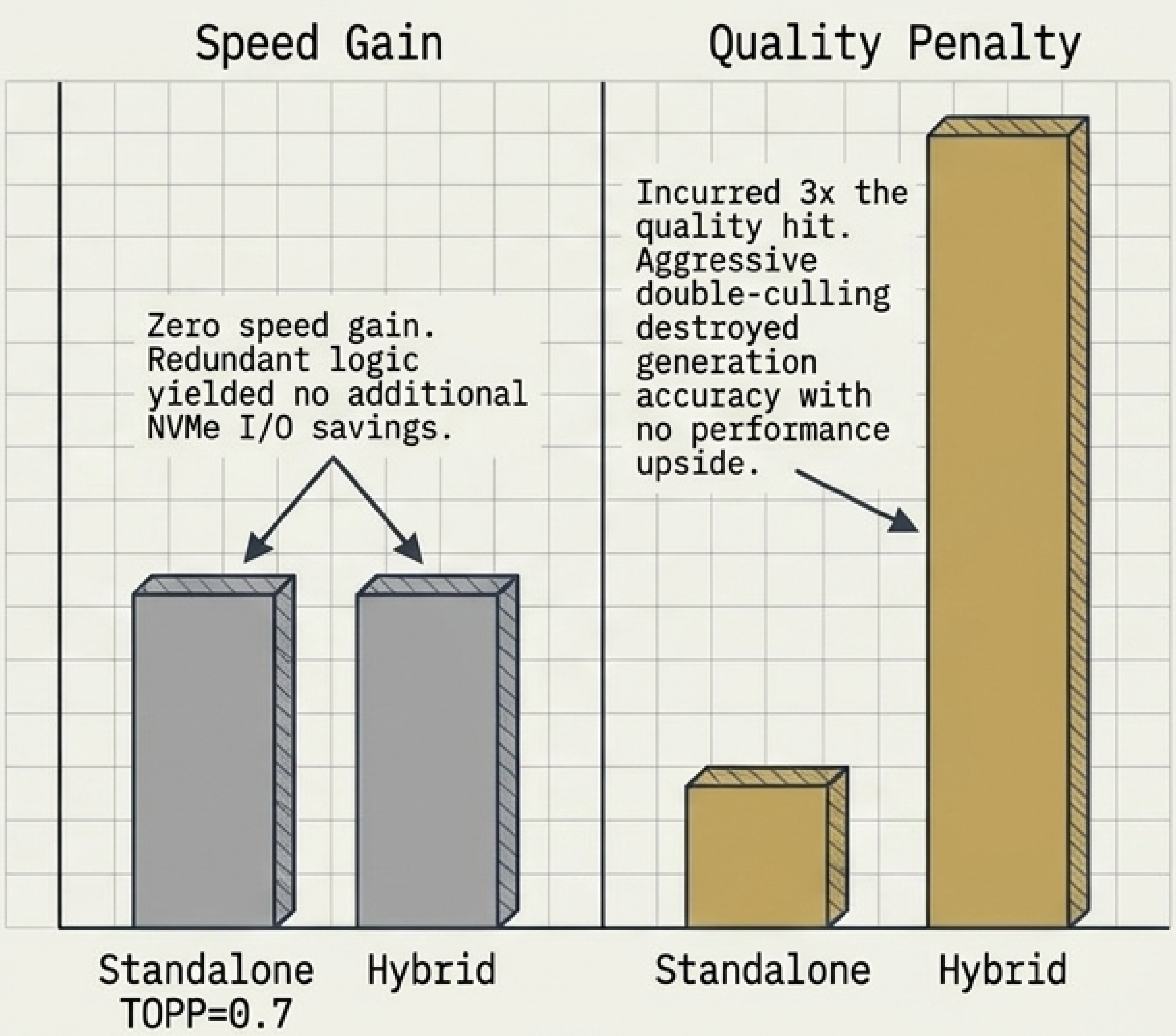


The Discarded Hybrid Approach

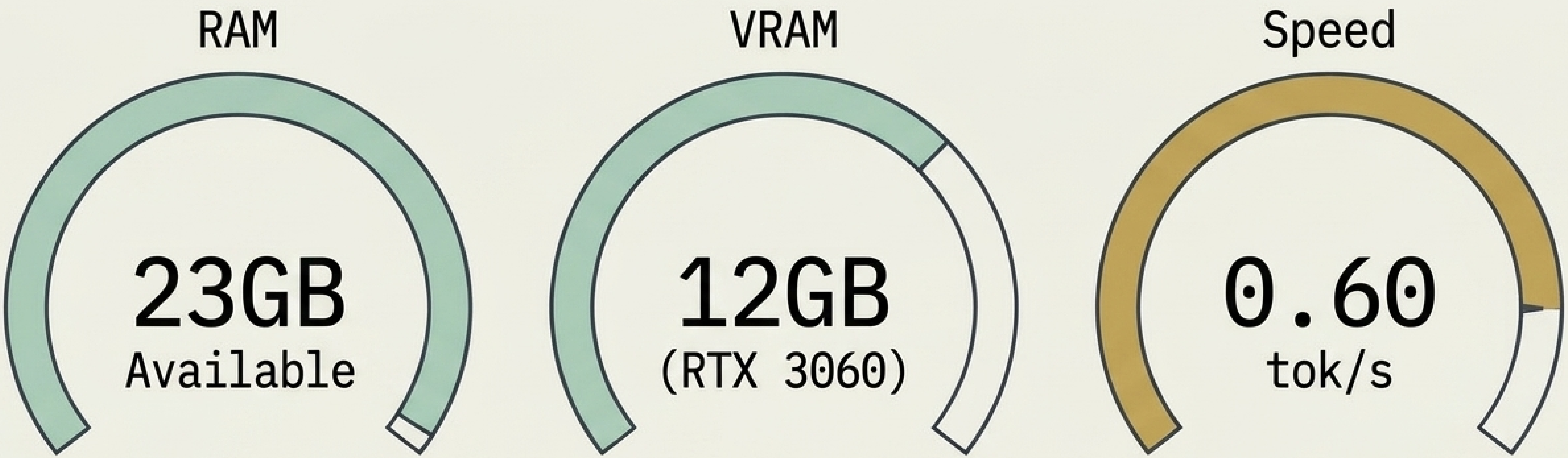
The Experiment

Combining top-K trimming with cumulative probability (top-P) trimming to aggressively cull experts (July findings 40.3). Formally discarded from the codebase.

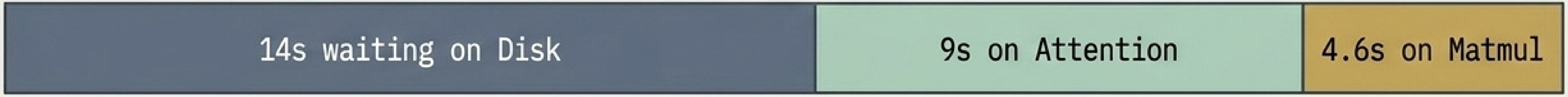
The Post-Mortem



Real-World Performance Limits (July 12 Record)



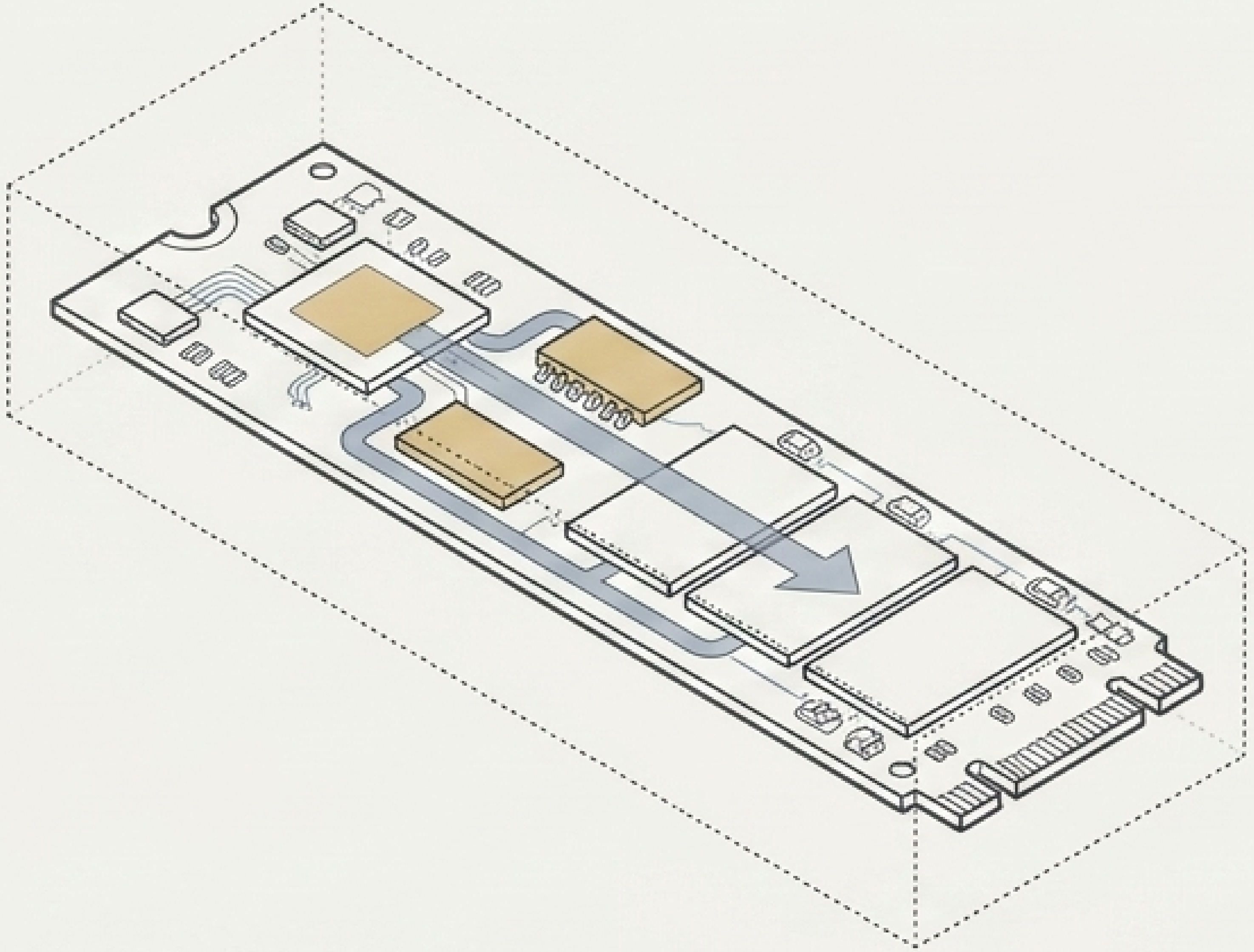
Execution Profile (cap=3, disk-bound)



Intel Xeon W-2140B (8C/16T) | WSL2 NVMe (ext4)

At 23GB RAM, the engine drops to cap=3 (holds ~3 experts/layer). The operation becomes completely disk-bound.

The Physics of SSD Endurance



Does pulling 4.7GB per token destroy a consumer SSD?

Peng-MIMO Load vs. Drive Wear Metric (TBW)

The Engineering Reality:

Consumer NVMe endurance is strictly rated in Terabytes Written (TBW).

Peng-MIMO generates an incredibly heavy read stream, which does not degrade TBW.

Thermal Warning:

While drive wear is safe, sustained multi-TB reads over long sessions may induce thermal throttling.

The Path to the 1.0 tok/s Barrier

Lever 1: Hardware Scaling

More RAM. The ultimate physical fix. Scaling to 32-64GB RAM allows the LRU cache to expand, eventually eventually reducing disk stalls to zero.

1.0 tok/s

Lever 3: Algorithmic

Native MTP. Integrated but currently off by default. Speculative drafting requires excess RAM for a warm expert cache to offset the heavy I/O cost of validation batches.

Lever 2: Environment Escapes

Native Linux. Removing the WSL2 VHDX virtualization overhead unleashes the raw, bare-metal O_DIRECT NVMe speeds.